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PALAEOMAGNETISM OF THE KORAS GROUP, NORTHERN CAPE PROVINCE, SOUTH AFRICA

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ABSTRACT

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Rocks of the Koras Group (ca. 1200—1000 Ma) near Upington, Namaqualand, mostly carry at least two components of NRM which have been distinguished by two stage a.f. plus thermal cleaning. Syngenetic or diagenetic NRM is identified in red sediments of the Kalkpunt Formation, in direction $D = 170°$, $I = -7°$ with $k = 67$, $\alpha_{95} = 7.4°$ corresponding to a palaeomagnetic pole at 57°S, 183°E ($d\psi = 4°$, $d\chi = 7°$). This result is corroborated by similar directions but opposite polarity in some immediately underlying basaltic andesites of the Florida Formation, and by soft components (anti-parallel to the hard component) in some specimens of both sediments and volcanics. NRM in the Ezelsfontein Formation, sampled at only one locality, may also be primary ($D = 29°$, $I = -11°$, $k = 54$, $\alpha_{95} = 16.9°$). Some of the basaltic andesites appear to have been completely remagnetized in an easterly direction $D = 95°$, $I = -10°$, $k = 55$, $\alpha_{95} = 10.4°$; the opposite polarity to this is found in soft components in some sedimentary specimens and both may reflect a mild magnetizing event at c. 800 Ma.

INTRODUCTION

Although the time interval ca. 1200—1000 Ma has been more intensively studied palaeomagnetically than much of the African Proterozoic, the pattern of apparent polar wander (a.p.w.) path during that time remains ill-defined. Palaeomagnetic data from the Koras Group, which outcrops in the vicinity of Upington in the north eastern Cape Province (Fig.1), can contribute to this problem because the time of formation of these strata (Table I) may span much of the interval in question. Moreover, preliminary studies by Briden (1975) showed that different parts of the Koras succession carried different stable NRMs, each of which, by comparison with the published a.p.w. path, appeared to correspond with different ages of magnetization within the late Proterozoic.

The Koras Group comprises an assemblage of unmetamorphosed sedimentary rocks, volcanics and volcaniclastics which were deposited in an intracratonic domain on a metamorphic basement which belongs to the marginal parts of the

mid-Proterozoic Namaqua mobile belt. The succession consists of five distinct lithostratigraphic units which are separated by angular unconformities. They are composed of clastic and shaly sediments, intercalated basaltic andesite, and two well-defined suites of quartz porphyry lavas with associated dykes. The total thickness of the Koras Group exceeds 7500 m, with the uppermost redbedtype sandstones and conglomerates making up ca. 3800 m of this pile. The Koras stratigraphy was established by Du Toit (1965) but his nomenclature has now been revised by the South African Committee for Stratigraphy (SACS), mainly on work by Vajner (1975). The old and new nomenclature are given in Table I. Stratigraphy and lithology suggest deposition in an unstable tectonic environment within graben-controlled basins along the so-called Doringberg lineament, a major fracture zone up to 60 km wide and over 300 km long (Vajner, 1975).

The lower units of the Koras Group are undated. The basaltic andesite of the Florida Formation yielded a 7-point Rb-Sr whole rock isochron with an age of 1178 ± Ma (Kröner et al., 1977) while the overlying quartz porphyry of the Leeuwdraai Formation gave a zircon Concordia intercept age of 1158 ± 98 Ma (Botha et al., in press). Granitoids genetically related to the upper Koras Group according to Geringer and Botha (1976) gave consistent ages of 1080—1050 Ma (Van Niekerk and Burger, 1967; Burger, in Geringer and Botha, 1976). There is a remarkable similarity in the depositional history, rock association and age between the Koras Group and the Sinclair Group farther northwest in Namibia, and Table I includes a suggested correlation of these sequences.

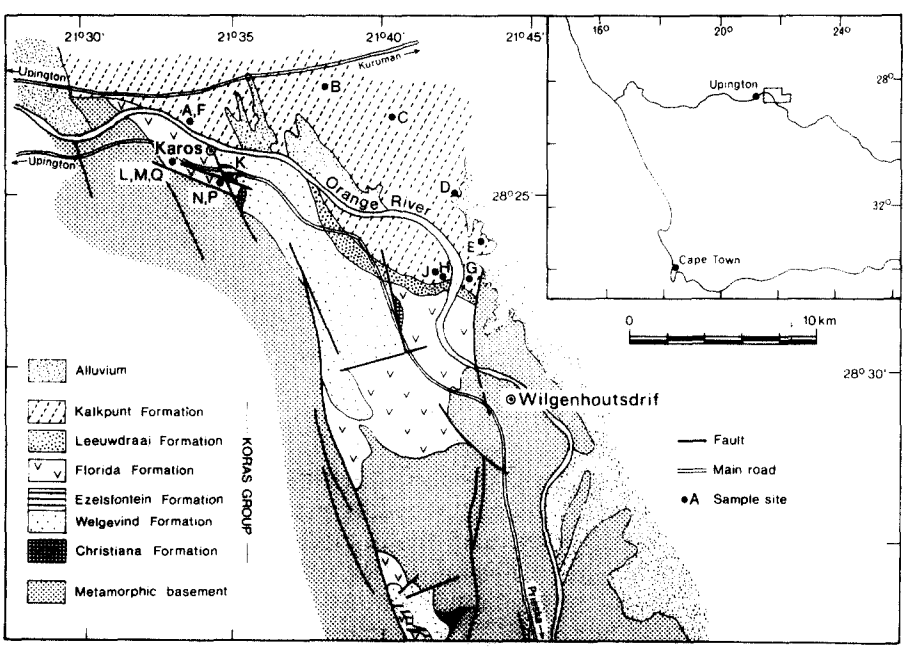


Fig. 1. Geological map, after Vajner (1975) showing palaeomagnetic sampling sites.

TABLE I

Lithostratigraphic subdivision of the Koras Group and proposed correlation with the Sinclair Group of Namibia

Koras Group			Sinclair Group	
old subdivision (Du Toit, 1965)	new subdivision (SACS)	age (Ma)	subdivision (Watters, 1977)	age (Ma)
Upper Sedimentary Group	Kalkpunt Formation	> 800 <1050	Auborus Formation	> 800 <1100
Unconformity			*Unconformity*	
Felsite dykes and granites		1080—1050 zircon		
Upper Quartz Porphyry	Leeuwdraai Formation	1158±98 zircon		
Unconformity				
Basaltic Andesite Group	Florida Formation	1178±18 Rb-Sr	Guperas Formation	> 960
Unconformity			*Unconformity*	
Middle Sedimentary Group	Ezelsfontein Formation			
Lower Quartz Porphyry	Welgevind Formation		Barby Formation	~1270
			Kunjas Formation	Rb—Sr
Unconformity				
	Christiana Formation		*Unconformity*	
			Nagatis Formation	~1290 Rb—Sr

Rb-Sr ages are based on decay constant $\lambda^{87}Rb = 1.39 \cdot 10^{-11} a^{-1}$, zircon ages are $^{207}/^{206}Pb$ minimum ages. Ages stated without method are inferred from field relationships.

Three 25-mm diameter cores about 100 mm long were drilled at each site, and oriented by sun compass. Samples from the Welgevind Formation and the dykes of the Leeuwdrai Formation proved to be unstable to a.f. demagnetization, and are not considered further here. One site (K) was sampled in the Ezelsfontein Formation; five sites were sampled in the Florida Formation (sites L—Q); and collections were made at nine sites in red siltstones and sandstones of the Kalkpunt Formation (A—J). The distribution of sampling sites is shown in Fig.1.

METHODS

Initial a.f. demagnetization revealed that many of the red sediment and basaltic andesite sites possessed at least two components of NRM with a softer component in some specimens at least two orders of magnitude larger than the high remanent-coercivity (H_{CR}) component. Thermal demagnetization of the a.f. cleaned specimens (i.e. with the soft component removed) showed that

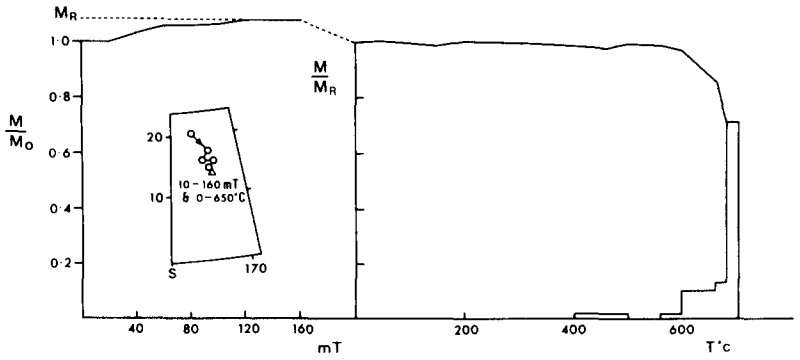
the high H_{CR} component was also characterized by critical blocking temperatures (T_B) greater than 600°C. This two-stage demagnetization procedure (Roy and Lapointe, 1978) determines whether the end direction isolated by a.f. cleaning is really single- or multi-component, and allows the high H_{CR}, high T_B component to be better isolated than by either demagnetization technique on its own.

KALKPUNT FORMATION

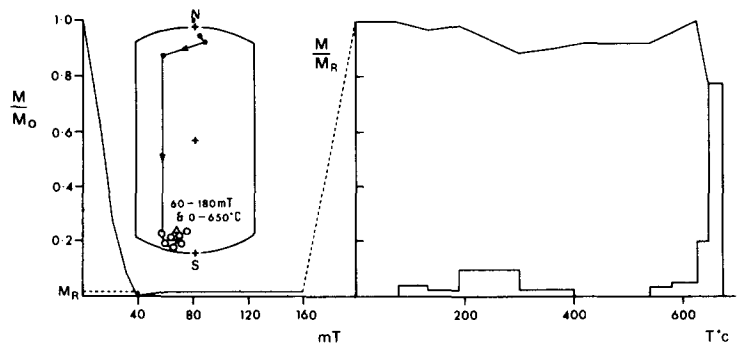
The intensities of total NRM for the red sediments were uncharacteristically high (up to about 0.5 Am^{-1}) and most site mean directions of total NRM were shallowly directed north or south (dip corrected) in oppositely polarised groups. Two stage demagnetization indicated that a component of NRM with high H_{CR}, and T_B greater than 600°C, is common to all red sediment sites (A—J). At three sites (A, H, J) this high H_{CR}, high T_B component is the only one present. A.F. demagnetization yields end-directions stable to peak fields up to 180 mT, close to the direction of total NRM, and NRM is virtually undemagnetized (Fig.2a). Further thermal demagnetization confirms that all this NRM is in one direction, and has blocking temperatures greater than 600°C (Fig.2a). Reflected light microscopy indicates that in these sites original (detrital?) magnetite has been almost completely pseudomorphed by hematite. Hematite in this form, together with discrete ilmenite grains and ultra-fine hematite (dust) comprises about 2% of these rocks.

In the remaining red sediment sites (B—G), two-stage analysis isolated two components of NRM. As well as the high blocking temperature component, another with low coercivities or blocking temperatures, and in varied directions, was distinguished. For sites C and D (Fig.2b) vector difference methods yield directions of the lower H_{CR} component (Fig.3) approximately antiparallel to the shallow, downward south-pointing direction of the high blocking-temperature component. Similar vector analysis of a.f. demagnetization trends for specimens from sites B and F fails to isolate the direction of the soft component owing to overlapping H_{CR} spectra with the high H_{CR}, T_B components.

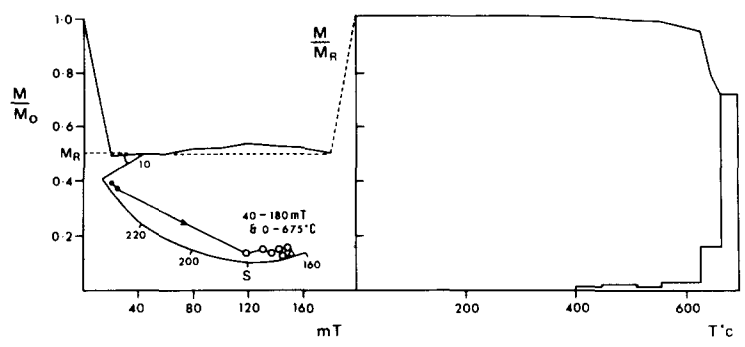
Fig.2. Examples of two-stage (a.f. followed by thermal) demagnetization of the Kalkpunt Formation (a, b, c), Ezelsfontein Formation (d) and Florida Formation (e), and a.f. demagnetization of the Florida Formation (f). In the thermal demagnetization diagrams (right hand side) the NRM intensity is normalized relative to the magnetization (M_R) residual after a.f. demagnetization (left hand side). The blocking temperature spectrum of M_R is shown below the respective decay curves. Inset diagrams show the directional changes of NRM (dip corrected) on a.f. demagnetization (dots or circles) and on further thermal demagnetization (triangles), and indicate the range of peak alternating fields and temperatures for which there is a single stable component of NRM. In Figs.2—4, open symbols denote upper hemisphere, closed symbols denote lower hemisphere on stereographic projections.



(a) A2b

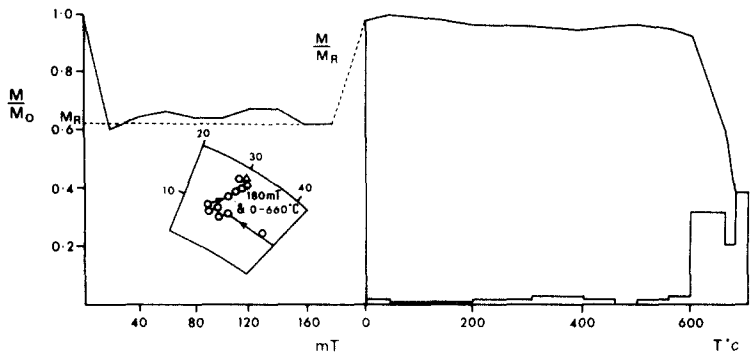


(b) D2a

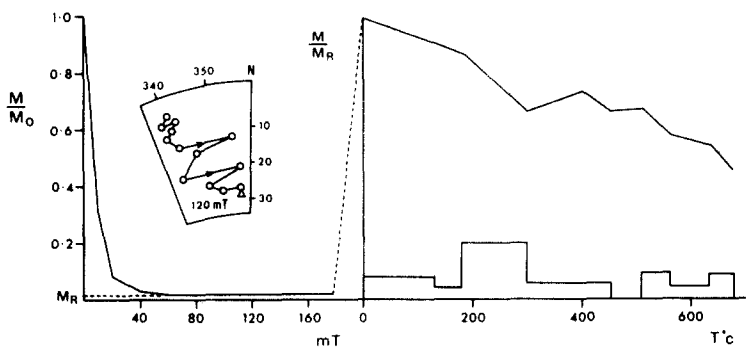


(c) E3c

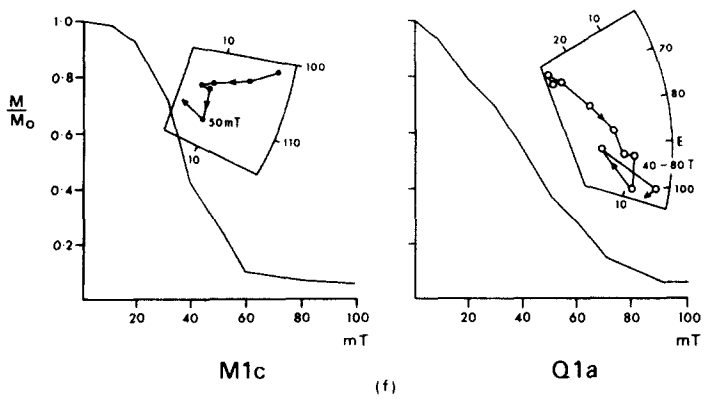
Fig. 2.



(d) K2



(e) P3a



(f)

Fig. 2 (cont.).

However the trend of the remagnetization planes for these specimens is consistent with the removal of a relatively soft shallow and northerly-directed component (Fig.3), i.e. like the soft component identified at C, D.

The presence of the two anti-parallel components may imply that individual specimens in sites C, D, B and F have recorded a field reversal, and the demagnetization results from site D most clearly illustrate this (Fig.2b). Demagnetization in peak fields less than about 60mT progressively removes the low H_{CR} component which, being two orders of magnitude larger than the high H_{CR} component, has a direction (Fig.3) close to the total NRM. The resultant NRM vector rotates through almost 180° (Fig.2b) and slightly increases in intensity above peak fields of 40mT, consistent with the preferential removal of the much larger soft component, relative to the anti-parallel high H_{CR} component.

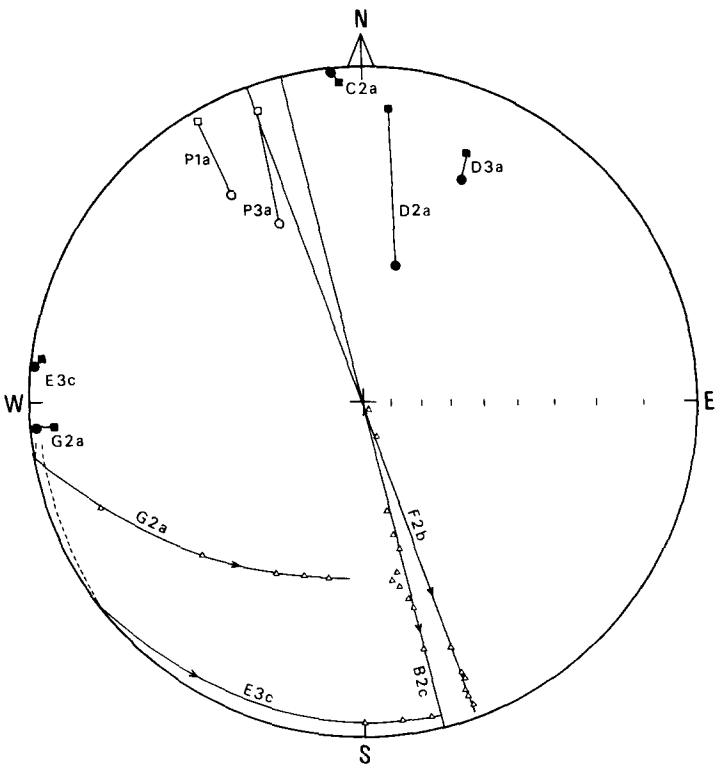


Fig.3. Directions of the low H_{CR} components determined by vector analysis of the two-component NRM in specimens from Kalkpunt Formation sites C and D (northerly) and sites E and G (westerly), and in Florida Formation site P. Dip corrected directions are denoted by squares; *in situ* directions are denoted by circles; *in situ* and dip corrected pairs are linked. Although the direction of the soft component in sites B and F of the Kalkpunt Formation could not be determined, the "remagnetization planes" for specimens in these sites are shown for comparison with the soft component directions in C and D.

Reflected light observations indicate that the single-component sites have magnetite completely pseudomorphed by hematite, whereas the sites with two components of NRM contain relict grains of magnetite, only partially pseudomorphed by hematite; quite fresh, unaltered cores of magnetite being mantled by hematite. This suggests that overprinting of NRM depended on the extent to which (probably detrital) magnetite was altered to hematite. This seems most likely to have taken place post-depositionally, although there is no direct evidence on this point.

Sites E (Fig.2c) and G also have two-component NRM, but the lower H_{CR} component (Fig.3) is shallow and westerly, which is quite different from the other two component sites.

EZELSFONTEIN FORMATION

Site K is very stable to a.f. demagnetization in peak fields above about 30 mT, and yields an end direction at peak fields of about 180 mT; magnetization retained after maximum a.f. demagnetization (M_R) has a single consistent direction during thermal demagnetization, and blocking temperatures are exclusively in the range 600—675°C (Fig.2d). Two-stage cleaning of all specimens from this site yields shallow upwards northeasterly NRM directions (Fig.4) both *in situ* and dip corrected; it is not clear whether directions should be dip corrected, because there is no indication of NRM age relative to folding.

Relict detrital magnetite grains are almost completely altered to hematite. Vestigial (111) cleavage attests the pseudomorphing of former magnetite.

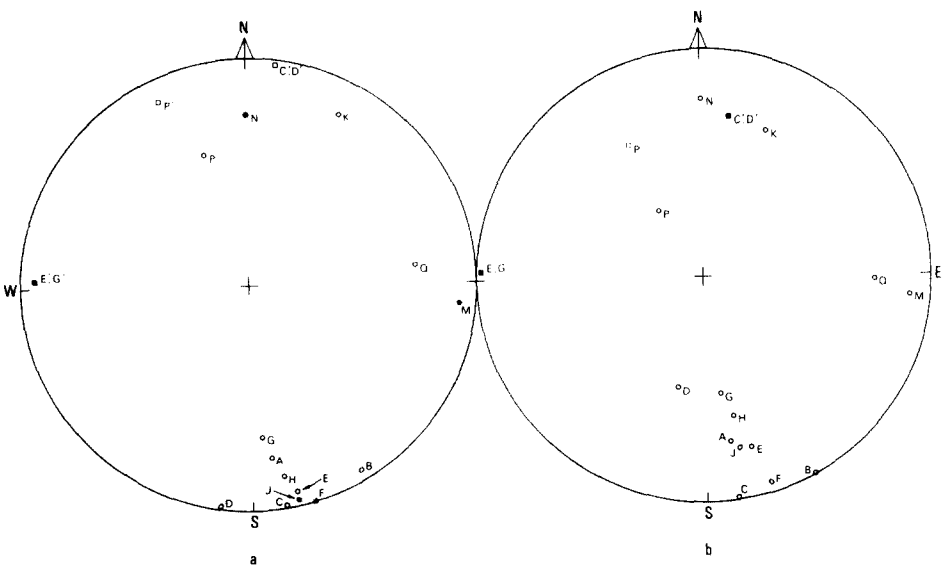


Fig.4. Site mean directions of NRM (○●); directions of low H_{CR}, low T_B components computed by vector difference methods are distinguished by squares and primes. (a) Dip corrected, (b) *in situ*.

FLORIDA FORMATION

The basaltic andesites of the Florida Formation (sites L—Q) are about as strongly magnetized as the strongest red sediment sites (ca. 0.5 Am^{-1}). Of these five sites, L was unstable against a.f. demagnetization. The bulk of NRM in the stable sites is demagnetized in peak fields from 40 mT to about 70 mT (Figs. 2e and f), and the asscoaited directional changes imply multi-component NRM. Two-stage cleaning (Fig.2e), however, indicates that the magnetization retained after maximum a.f. demagnetization (M_R) at 160 mT has only one component, and this is in a consistent direction close to the end-direction of a.f. demagnetization, and has distributed blocking temperatures.

In situ directions of the low H_{CR} component determined for site P by vector analysis (Fig.3) are very similar to the dip-corrected end-directions of a.f. demagnetization, which suggests that NRM was acquired in site P both shortly before and shortly after folding of the lavas.

Two-stage cleaning of all the andesite sites shows that those from the Florida andesite quarry (M,Q) have shallow easterly directions (Fig. 4a) whereas those from the road cut at Karos (N and P) have northerly directions. As for the Ezelsfontein Formation, it is not clear whether *in situ* NRM directions should be dip corrected. The main Fe-Ti oxides in the basaltic andesites are skeletal titanomagnetites with hematite lamellae and large ilmenite phenocrysts.

INTERPRETATION OF NRM DIRECTIONS

Fisher precision k within samples ranges from 14 to 1255. When unit weight is given to the samples, k within sites ranges from 65 to 1723. Because within-sample dispersion is thus a major contributor to overall dispersion and because there are only about three specimens per sample and three samples per site — inadequate for a site mean calculation on a two-tier basis (Watson and Irving, 1957) — the site means are calculated by giving unit weight to specimens (Table II and Fig.4). The corresponding poles (Fig.5) are distributed in a general way along the African a.p.w. path for the time interval 1200—600 Ma.

The virtual geomagnetic pole (VGP) from the Ezelsfontein Formation falls close to the published path at ca. 1200 Ma; this fit, and the separation of nearly 30° between it and the poles from the Florida Formation site N and Kalkpunt Formation sites D and G might lend support to the view that unconformities above the Ezelsfontein Formation represent as much as 100 Ma hiatus (Table I). However the Ezelsfontein data came only from a single site and may not be fully representative, so too much weight should not be ascribed to this conclusion.

There are several observations which indicate that the red sediments of the Kalkpunt Formation and the basaltic andesite of the Florida Formation at sites N and P acquired their magnetization during deposition and/or diagenesis.

(1) The direction of stable NRM in these basaltic andesites is reversed relative to the high T_B, high H_{CR} component in the overlying sediments, the antiparallelism being most precise for N relative to A and G.

TABLE II

Site statistics of a.f. and thermally cleaned NRM

Site	Treatment (mT/°C)	N	n	R	k	α_{95}	In situ				Dip-corrected			
							D	I	VGP		D	I	VGP	
									Lat.(N)	long. (E)			lat. (N)	long.(E)
Kalkpunt Formation														
A	180/—	3	7	6.9831	354	3.2	172	−18	−52	189	174	−15	−54	191
B	180/450	2*2	6	5.7571	21	15.1	151	0	−50	152	150	− 3	−49	153
C	180/450	3	8	7.8566	49	8.0	172	− 1	−60	185	171	− 1	−60	184
D	180/—	3	6	5.9048	53	9.3	194	−37	−39	218	188	− 1	−60	218
E	180/450	3	10	9.9543	197	3.4	166	−15	−51	179	168	− 4	−57	178
F	180/450	3	7	6.9610	154	4.9	163	− 3	−56	171	164	0	−57	170
G	180/450	3	11	10.8558	69	5.5	173	−35	−42	193	176	−23	−49	195
H	180/450	3	7	6.8639	44	9.2	170	−26	−52	188	171	− 9	−56	186
J	180/—	3	8	7.9684	221	3.7	169	−15	−52	183	168	2	−60	177
C/D*1	<100/—	3	3	2.7873	9	—	10	20	50	37	8	− 2	62	39
E/G*1	< 50/—	2	2	1.9881	84	—	272	1	1	293	272	+ 4	1	294
Florida Formation														
M	30/—	3	7	6.8703	46	9.0	96	− 6	− 4	117	96	+ 5	− 7	113
N	30/—	3*2	8	7.9129	80	6.2	0	−15	69	22	0	+17	53	22
P	140/—	3	8	7.7163	25	11.4	326	−52	61	275	342	−28	68	327
Q	50/—	2*2	4	3.9035	31	16.7	92	−17	3	120	84	−18	10	117
P*1	< 70/—	2	2	1.9860	71	—	331	−24	59	316	335	− 7	56	334
Ezelsfontein Formation														
K	180/—	3	8	7.8329	42	8.7	24	−21	61	79	29	−10	54	77

*1 Soft component.

*2 Third sample excluded as unstable. *n* is the number of specimens; the notation is otherwise standard (Fisher 1953)

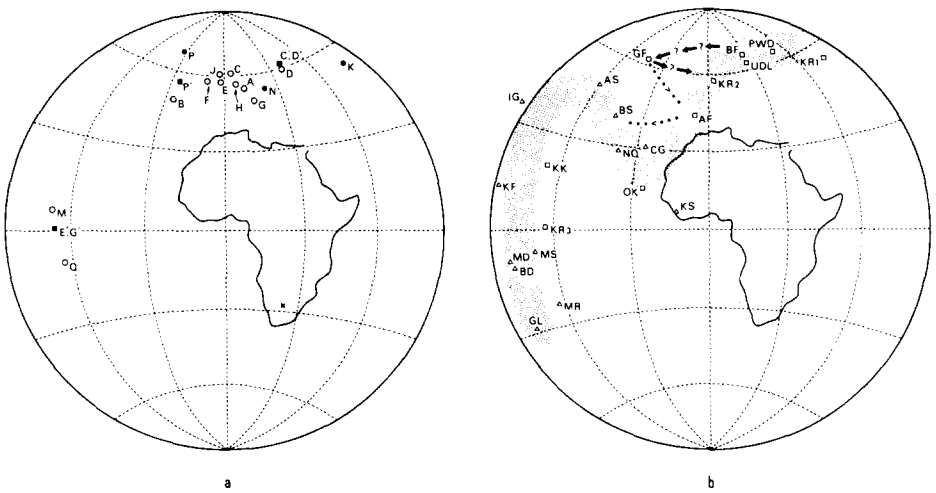


Fig.5. (a) Virtual geomagnetic poles for sites from the Koras Group (Table II). Area studied is denoted by a cross. (b) Apparent polar wander path for Africa between about 1200 and 800 Ma: swathe after McElhinny and McWilliams (1977); dotted line after Piper (1976); strong arrows indicate revision suggested in this paper. Equal area equatorial projection; squares denote Kalahari craton and triangles denote Congo craton. AF = Auborus Formation, AS = Abercorn Sandstone, BD = Bukoban dolerites, BF = Barby Formation, BS = Bukoba Sandstone, CG = Chela Group, GF = Guperas Formation, GL = Gagwe Lavas, IG = Ikorongo Group, KF = Kigonero Flags, KK = Klein Karas dykes, KR1, 2, 3 = Koras Group (entries 6, 3 and 5, respectively, of Table III of this paper), KS = Kisii Series, MD = Mbala dolerites, MR = Manyovu Redbeds, MS = Malagarasi Sandstone, NQ = Nosib Group quartzites, OK = O'okiep intrusives, PWD = post-Waterberg diabases, UDL = Umkondo dolerites and lavas.

(2) The single high T_B, high H_{CR} component at sites A, H and J is indistinguishable from the higher T_B component at sites in the Kalkpunt Formation which carry two-component NRM. There is no evidence of discrete post-diagenetic episodes of chemical alteration of the Koras sediments which might have led to chemical remagnetization.

(3) The soft component at sites C and D and possibly also B and F is antiparallel to the hard component at those sites. The correlation of the lower coercivity component with vestigial magnetite and the higher coercivity component with the hematite alteration product indicates that (contrary to what is usually and simplistically assumed in palaeomagnetism) the softer component is the older. Hence observations 1—3 are most easily explained in terms of polarity reversal (N—S with respect to the a.p.w. path illustrated in Fig.5) with the N interval being recorded in the basaltic andesites and in the *softer* component in the sediments, and the S interval by the harder component in the sediments. If this is correct then the stable NRMs predate tilting of the rocks, and the dip-corrected NRM directions are the best estimates of the local palaeomagnetic field when the Koras rocks were formed.

(4) Blocking curves for hematite (Pullaiah et al., 1975) preclude thermal remagnetization of the high blocking temperature component in Kalkpunt red

sediments, since there is no evidence that the Koras Group has been affected by the temperatures necessary for such remagnetization.

(5) Red sediments at sites E and G have a soft magnetization (Fig.3) almost at right angles to their hard component, and which is interpretable as being of ca. 800 Ma age (see below). In contrast to the remainder of the Kalkpunt Formation, it is thus suggested that the soft component is several hundred million years younger than the hard component, which would be consistent with its association with a mild thermal event.

The site mean poles of stable NRM in the Kalkpunt Formation are in accord with the result from a single site previously published by Briden (1975). They are spread in an east—west direction, in no particular stratigraphic order except that the more westerly poles tend to come from the higher parts of the sequence. Because this distribution is not Fisherian, it is not obvious how the data should be summarized, so several alternative combinations are presented in Table III. They all have the same overall mean within 5°, and the same circle of confidence within the range 8.5° ± 1.2, so it matters little which is adopted. They all have about double the precision for dip-corrected means than for *in situ* directions. Although in no cases is this conclusive evidence with 95% confidence that the remanence predates the folding, it is strongly indicative that this is likely to be so with confidence approaching 90%. The first way of combining the data is simply to take the overall mean of all sites, and to quote

TABLE III

Palaeomagnetic summary statistics

		In situ			
	S^{*1}	R	k	k_u, k_l	α_{95}
(1) Kalkpunt Formation, all sites simple dip-corrected	9	8.6391	22	39.6, 9.5	11.2
(2) As 1, plus Florida Formation, site N reversed	10	9.4909	18	31.5, 8.2	11.8
(3) As 1, minus sites B, D	7	6.8591	43	83.6, 15.8	9.4
(4) Florida Formation, sites M, Q	5 (samples)	4.9269	55		10.4
(5) As 4, plus soft component E/G reversed and treated as one site	3	2.9762	84		13.5
(6) Ezelsfontein Formation, site K, dip-corrected	3 (samples)	2.9631	54		16.9

k_u, k_l are upper and lower 95% confidence limits on k (Cox, 1969).
*1 Number of sites, unless otherwise stated.
*2 Pole computed from dip-corrected RM direction except for entries 4 and 5.

Fisher errors despite their not being strictly applicable. Florida Formation site N could be added to this group, as could site P. Alternatively B and D could be excluded from the remaining sites in the Kalkpunt Formation on the grounds that their means are statistically different from the remainder with more than 99% probability (Watson 1956); the rest of the Kalkpunt site mean directions do then form a Fisherian set. For summary purposes this last combination is the most conservative and rigorous, and is the one which is quoted in the abstract and illustrated as pole KR2 in Fig.5b, though its use in isolation would ignore the data from the Ezelsfontein and Florida Formations.

These VGPs (Fig.5a) are all close to published poles for the time interval ca. 1200—1050 Ma (c.f. Figs. 5a and b) through which two published alternative a.p.w. paths (Piper, 1976; McElhinny and McWilliams, 1977) have already been drawn. The single VGP from the Ezelsfontein Formation (entry 6 of Table III; pole KR1 of Fig. 5b) lies close to the beginning of this path, consistent with its magnetization being syngenetic or early diagenetic. The mean palaeomagnetic pole KR2 (Fig.5b) from the Kalkpunt and Florida Formations (entry 3 in Table III) lies between the mean poles from the Barby and Auborus Formations of the Sinclair Group, Namibia, and this is consistent with their stratigraphic correlation (Table I). The spreading of the Koras VGPs in an E—W direction (Fig.5a) extending close to the published pole from the Guperas Formation, suggests that the magnetization of the Guperas is of comparable

		Dip-corrected						Palaeomagnetic pole*[a]		
D	I	R	k	k_u, k_l	α_{95}	D	I	lat.(N)	long.(E)	$d\psi$ $d\chi$
169	−17	8.7956	39	70.3,16.8	8.3	170	−6	−57	183	4,8
171	−14	9.7655	38	66.5,17.4	7.9	171	−7	−57	185	4,8
169	−16	6.9109	67	130.3,24.6	7.4	170	−7	−57	183	4,7
95	−10	4.8460	26		15.3	92	−4	−2	118	—
93	− 8	2.9486	39		20.0	91	−6	−1	117	7,14
23	−22	2.9631	54		16.9	29	−11	55	77	—

age to that of the Koras. The scatter in the palaeomagnetic data, uncertainties as to how soon after deposition the stable NRM originated, and limitations of lithological correlation, make it uncertain whether the Kalkpunt Formation NRM is older than that of the Guperas Formation (and falls *before* the Guperas pole on Piper's (1976) version of this part of the a.p.w. path) or younger. This second alternative, illustrated in Fig.5b is, however, more consistent with the stratigraphy of the various rock units, and in the absence of sure indication that sequence of *magnetization* was different from sequence of *formation* we now regard it as the most probable a.p.w. path for ca. 1200—1050 Ma. This path differs from that of McElhinny and McWilliams (1977) in taking account of the results from the Guperas Formation (which they excluded on mistaken geochronological reasoning) and the Barby Formation (whose magnetization age we contend they overestimated). It follows McElhinny and McWilliams (1977) and not Piper (1976) in taking account of the well established result from the O'okiep intrusions.

Briden (1975) concluded that NRM in the Florida Formation at Karos (his sites 7—9) lay on the pre-1200-Ma portion of the a.p.w. path and hence predated the NRM in the overlying sediments. The new data contradict this by showing that NRM in basaltic andesites M, Q is softer than that in petrographically similar samples at sites N and P and is therefore likely to be younger. Moreover the soft components in red beds E, G referred to above, correspond to very similar VGPs, though of opposite polarity. All these are close to the published a.p.w. path just before the poles from the Bukoban dolerites (which are $\geqslant$ 800 Ma old). Since it appears that these magnetizations postdate the deformation of their host rocks, it is their *in situ* remanences which are meaningful and are translated into VGPs in Fig.5. It is concluded that the Florida Formation sites M and Q were totally remagnetized at about that time while the sediments E and G were only partially remagnetised. The sediments presumably escaped complete remagnetization because their NRM, unlike that of the andesites, is characterized by very high T_B and H_{CR}. We suggest that this local and variable effect can be correlated with widespread pegmatitic activity associated with the Namaqua metamorphic complex, dated in this area between ca. 960 and 830 Ma largely by U—Pb on zircons, apatite and monazite (Burger and Coertze, 1973). That an event around that time affected the Koras rocks themselves is evidenced by a single $^{39}Ar/^{40}Ar$ age determination on a basaltic andesite of the Florida Formation at 794 Ma, reported by Burger and Coertze (1975) who explain it in terms of overprinting.

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